Parusan Natheeswaran

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Summary of Qualifications

- Computer Engineering co-op student who builds and ships complete systems, from offline language-research software and AI-powered web products to formally verified RISC-V CPU cores.
- Combines applied AI with rigorous engineering: typed parsers, local model orchestration, tool-calling agents, reproducible experiments, automated testing, and CI/CD.
- Works across disciplines: delivered player-facing features with design, engineering, and QA at Riot Games, develops speech and conversational AI at Outlier, and leads 10+ staff per shift.

Education

Toronto Metropolitan University **Sep. 2023 – Present**
Bachelor of Engineering, Computer Engineering (Co-op) *Toronto, ON*

- Co-op Availability: January 2027 (12 months) or May 2027 (8, 12, or 16 months)
- Relevant Coursework: Software Systems, Digital Systems, Operating Systems, Algorithms & Data Structures

Technical Skills

Languages: Python, TypeScript, JavaScript, C, C++, Java, Go, SQL, SystemVerilog, Bash
AI/ML: PyTorch, AMD ROCm, LLMs, AI agents, tool calling, RAG, QLoRA fine-tuning, Claude API, Ollama, whisper.cpp
Software & Cloud: React, Electron, Node.js, Flask, FastAPI, REST APIs, PostgreSQL, Redis, Docker, AWS, Azure, Fly.io
Hardware: RISC-V, CPU pipelines, FPGA, formal verification (SymbiYosys), cocotb, Verilator, Yosys/nextpnr
Tools & Process: Git, GitHub Actions CI/CD, pytest, Vitest, Playwright, Linux, JIRA, Agile Scrum, code reviews, Figma

Work Experience

Shift Manager **Dec. 2025 – Present**
Town of Milton *Milton, ON*

- Lead 10+ staff per shift, handling escalations, safety oversight, process compliance, and weekly status reports.

Machine Learning Engineer **Oct. 2024 – Present**
Outlier AI *Oakland, CA (Remote)*

- Improved accuracy and reliability across 20+ AI/ML projects by developing and optimizing speech recognition and conversational AI models in Python, C++, and Java.
- Earned 70+ acknowledgments by owning end-to-end workflows, from exploratory data analysis to model deployment.

Game Developer **May 2024 – Aug. 2024**
Riot Games *Los Angeles, CA (Remote)*

- Shipped a playable character to millions of users on a 3-month deadline, prototyping features in Python.
- Cut crash rates 15% by triaging and resolving 50+ critical bugs with design, engineering, and QA teams.

Projects

Xenolinguist | *TypeScript, React, Electron, Node.js, Ollama* | *Download RC* **Apr. 2026 – Present**

- Built a linguistic research engine with typed parsing, grounded grammar induction, arithmetic grammars and active question selection; achieved 150/150 withheld meanings across 30 synthetic languages with a supplied grammar prior.
- Engineered reviewable local-AI proposals, native audio provenance, revisioned saves and crash recovery; shipped an unsigned Windows v1.1.0-rc.1 prerelease with 529 unit tests and 56 workbench browser checks per platform.
- Verified fresh installation and upgrades against 3,779 packaged files and nine archive round trips; published reproducible benchmarks, retained failure cases and a three-minute demonstration of an unseen symbolic derivation.

Bingery | *React, TypeScript, Python, Flask, Claude API, Fly.io* | *Live Demo* **Mar. 2026 – Present**

- Built a live anime discovery PWA with a Claude tool-calling AI agent (6 custom tools, cloud or local models), an IDF-weighted similarity engine, and a 75-route JWT-secured REST API.
- Backed it with 690+ TDD-first tests in GitHub Actions CI/CD, daily jobs syncing a 14,000+ title catalog, and a mobile rework driven by usability testing with 15+ users.

Pulsar-V | *SystemVerilog, RISC-V, FPGA, cocotb, SymbiYosys, Yosys/nextpnr* **Jul. 2026 – Present**

- Designed two RV32I CPU cores, including a 5-stage pipeline with forwarding and a BTB branch predictor (83.8–99.6% accuracy, 1.00–1.27 CPI), closing timing at 38.8 MHz on a Lattice ECP5 FPGA.
- Verified both cores with 43/43 riscv-formal proofs, 38/38 riscv-arch-test against the SAIL model, and 41/41 riscv-tests.

EdgeSpark | *Python, PyTorch, AMD ROCm, Radeon RX 7900 XTX* **Apr. 2026 – Present**

- Speculative decoding on one AMD GPU: exact Qwen3-4B output, 9.5 GB VRAM, +43% modelled throughput.